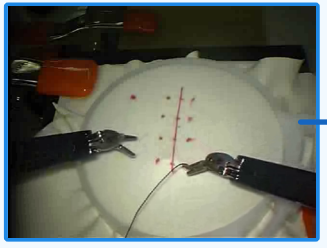
Decoupled Multimodal Architecture: Frozen Surgical Video Foundation Features + Kinematics with Adaptive Reliability Gating

OPTICAL PERCEPTION STREAM (Endoscopic Video)



JIGSAWS Video Frame
640×480 @ 30 FPS

Frozen SurgMotion
ViT-Large / ResNet-18

Video-JEPA Pretrained
(3,658h Surgical Video)

$\mathbf{z}_t^{\text{vis}} \in \mathbb{R}^{1024}$

Video Temporal Head
2-Layer GRU (H=128)

Proj: 1024D → 128D
Window S=8 (16 frames)

$\mathbf{h}_t^{\text{vis}} \in \mathbb{R}^{128},\ \mathbf{y}_t^{\text{vis}} \in \mathbb{R}^C$

ROBOT TELEMETRY STREAM (da Vinci Kinematics)

76-D da Vinci
Robot Telemetry

- Master L/R (38D)
- Slave L/R (38D)
- XYZ, Rot R(3×3)
- Lin/Ang Vel, Jaw θ

Leakage-Safe
Z-Score Normalizer

Stats (μ_tr, σ_tr) computed
strictly on training trials
Zero test leakage

$\hat{\mathbf{k}}_t \in \mathbb{R}^{76}$

Kinematics Head
2-Layer GRU (H=128)

Proj: 76D → 128D
LayerNorm, Dropout=0.2

$\mathbf{h}_t^{\text{kin}} \in \mathbb{R}^{128},\ \mathbf{y}_t^{\text{kin}} \in \mathbb{R}^C$

Availability Vector

$\mathbf{a}_t = [a_t^{\text{vis}}, a_t^{\text{kin}}]^\top \in \{0,1\}^2$

RELIABILITY-AWARE
DYNAMIC GATING

Input : $[\mathbf{h}_t^{\text{vis}}; \mathbf{h}_t^{\text{kin}}; \mathbf{a}_t]$

$\tilde{\boldsymbol{\alpha}}_t = \text{Softplus}(\mathbf{W}_g \mathbf{x} + \mathbf{b}_g)$

$\alpha_t^m = \frac{\tilde{\alpha}_t^m a_t^m}{\sum_l \tilde{\alpha}_t^l a_t^l + \varepsilon}$

$\tilde{\alpha}_k \approx 80\% - 95\%$
$\tilde{\alpha}_v \approx 5\% - 20\%$

Fallback Guarantee :
$a_t^{\text{vis}} = 0 \rightarrow \alpha_t^{\text{vis}} \equiv 0, \alpha_t^{\text{kin}} \equiv 1$
$\hat{\mathbf{y}}_t = \mathbf{y}_t^{\text{kin}}$ (exact)

DYNAMIC LOGIT
FUSION

$\hat{\mathbf{y}}_t = \alpha_t^{\text{vis}} \mathbf{y}_t^{\text{vis}} + \alpha_t^{\text{kin}} \mathbf{y}_t^{\text{kin}}$

Softmax Classifier
$\hat{g}_t = \arg\max_c P(g_t = c)$

Gesture ĝ_t ∈ {G1..GC}